

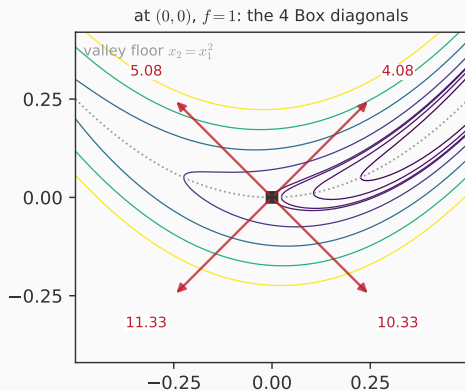
Optimization

3.2.5 Hooke–Jeeves method (pattern search)

José Firmino Aguilar Madeira

What if, instead of the diagonals, we try the axes?

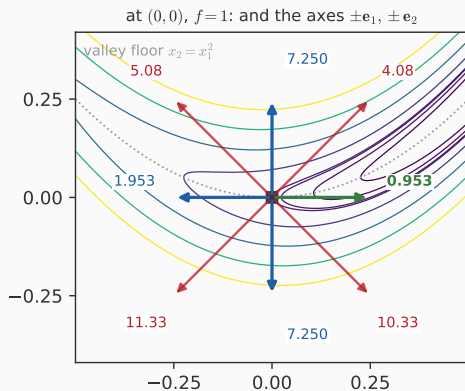
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But along the x_1 axis: $f(0.25, 0) = 0.953 < 1$.



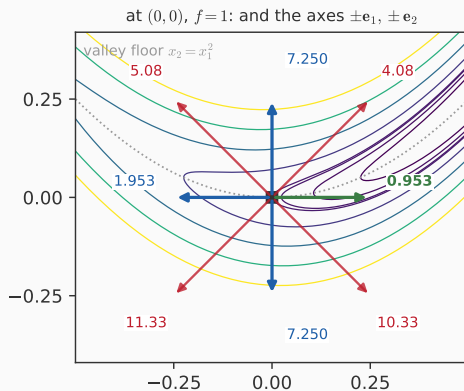
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Why not test $\pm \mathbf{e}_1, \dots, \pm \mathbf{e}_n$?

n	Box: 2^n	axes: $2n$
2	4	4
10	1 024	20
20	$\approx 10^6$	40



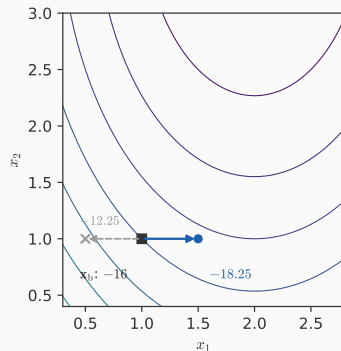
One exploration by hand

$f(x_1, x_2) = 3x_1^2 + x_2^2 - 12x_1 - 8x_2$ (minimum at $(2, 4)$, $f = -28$).

Base point $\mathbf{x}_b = (1, 1)$, $f = -16$; perturbations

$\mathbf{P} = (0.5, 0.5)$.

1. **Coordinate** x_1 : $(1.5, 1) \rightarrow -18.25$,
 $(0.5, 1) \rightarrow -12.25$: keep the best, $(1.5, 1) \checkmark$



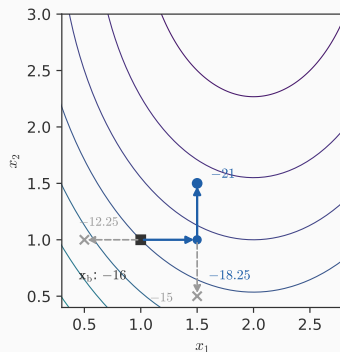
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 $(1.5, 1.5) \rightarrow -21$, $(1.5, 0.5) \rightarrow -15$: keep $(1.5, 1.5) \checkmark$



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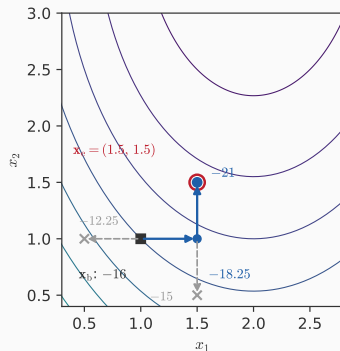
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3. $\mathbf{x}_b = (1, 1) \rightarrow \mathbf{x}_e = (1.5, 1.5)$, with 4 evaluations.

Coordinates explored **sequentially**, each from the **best point so far**; **both directions** are tried and the best is kept (Deb, 2012).



Exploration around $\mathbf{x}_b$ with perturbations

$\mathbf{P} = (P_1, \dots, P_n)$

```
1:  $\mathbf{x} \leftarrow \mathbf{x}_b$ ;  $f_{\text{best}} \leftarrow f(\mathbf{x}_b)$ 
2: for  $j = 1, \dots, n$  do
3:   evaluate  $x_j + P_j$  and  $x_j - P_j$ 
4:   keep the best of the three ( $\mathbf{x}$  and the two trials)
5:   (if neither improves  $f_{\text{best}}$ ,  $x_j$  stays as it was)
6: end for
7: return  $\mathbf{x}_e \leftarrow \mathbf{x}$  ▷ failed if  $\mathbf{x}_e = \mathbf{x}_b$ 
```

$n_f = 2n$ per exploration

- Both directions are always tried (Deb, 2012): $n_f = 2n$, but the **best** direction is kept, not the first one that improves.
- **Box**: **all combinations** of $\pm (2^n)$.
Hooke–Jeeves: **one coordinate at a time** ($2n$).
- The direction obtained, $\mathbf{x}_e - \mathbf{x}_b$, has components in $\{-P_j, 0, +P_j\}$: a rough indication of the direction of progress (not the best one), without derivatives.

We have a direction that worked: why not extrapolate?

The exploration took us from $\mathbf{x}_b$ to $\mathbf{x}_e$ and improved. We have information that Box did not exploit:

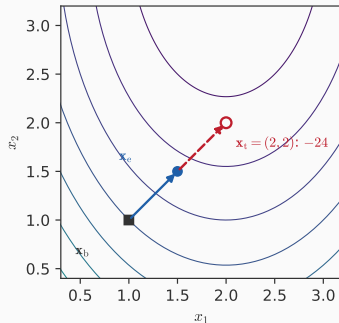
$$\mathbf{d} = \mathbf{x}_e - \mathbf{x}_b.$$

Why explore again only around $\mathbf{x}_e$? If that direction worked:

we go further in the same direction

$$\mathbf{x}_t = \mathbf{x}_b + a(\mathbf{x}_e - \mathbf{x}_b), \quad a = 2: \quad \mathbf{x}_t = 2\mathbf{x}_e - \mathbf{x}_b.$$

In the example: $\mathbf{x}_t = 2(1.5, 1.5) - (1, 1) = (2, 2)$, $f = -24$.



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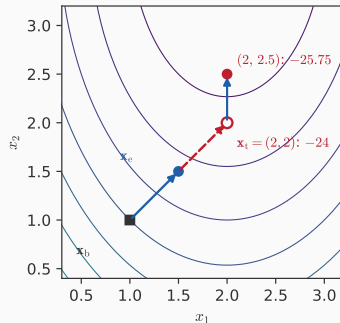
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And we **explore again** around $\mathbf{x}_t$: $(2, 2) \rightarrow (2, 2.5)$,
 $f = -25.75 < -21$ — the jump is accepted.



Exploratory + pattern: the Hooke–Jeeves method

Hooke and Jeeves (1961): the two moves we have just built, alternated.

Exploratory move

Local search, axis by axis, from the base point $\mathbf{x}_b$: perturb each coordinate by $\pm P_j$ and keep whatever improves. Result: the explored point $\mathbf{x}_e$. Cost: $2n$.

Pattern move

Extrapolation in the direction of progress: trial point $\mathbf{x}_t = \mathbf{x}_b + a(\mathbf{x}_e - \mathbf{x}_b)$, followed by a new exploration. Repeated while there is progress: the jumps *accumulate* the direction that works.

the pattern **accelerates**

the exploration **corrects**

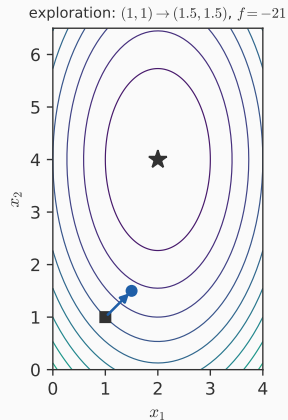
The exploratory steps are aligned with the axes, but extrapolation allows following oblique valleys.

The perturbations P_j are the *poll step*: they play the role of the step α_k of the other methods (and return in 5.1 and 6.5).

Complete example: the path curves

$\mathbf{x}_0 = (1, 1)$, $a = 2$, $\mathbf{P}_0 = (0.5, 0.5)$, $\mathbf{T} = (0.1, 0.1)$ (large tolerance, just so the example ends quickly).

1. **Exploration** at $(1, 1)$, $f = -16$: $(1.5, 1) \rightarrow -18.25$ ✓; $(1.5, 1.5) \rightarrow -21$ ✓ (the best of each pair). $\mathbf{x}_e = (1.5, 1.5)$.

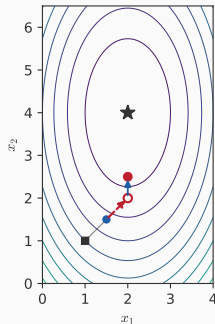


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2. **Pattern**: trial $2(1.5, 1.5) - (1, 1) = (2, 2)$, $f = -24$; exploration $\rightarrow (2, 2.5)$, $f = -25.75 < -21$: accepted.

pattern $(2, 2)$, explor. $\rightarrow (2, 2.5)$, $f = -25.75$: accepted

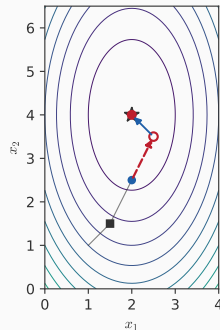


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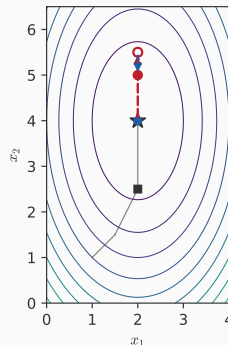


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4. **Pattern**: trial $2(2, 4) - (2, 2.5) = (2, 5.5)$, $f = -25.75$; exploration $\rightarrow (2, 5)$, $f = -27 \geq -28$: **rejected**. Back to $(2, 4)$.

pattern $(2, 5.5)$, explor. $\rightarrow (2, 5)$, $f = -27$: rejected

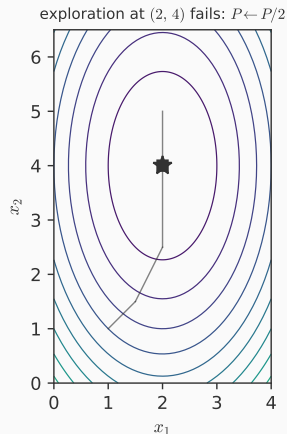


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5. **Exploration** at $(2, 4)$ fails with $P = 0.5$; with 0.25 ; with 0.125 ; $0.0625 < T$: **stop**. Solution $(2, 4)$, $f = -28$.

The pattern always points in a straight line; the exploration corrects; the path **curves**.



The algorithm

explore around x_b
fails $\Rightarrow P \leftarrow P/2$
improves $\Rightarrow$ **extrapolate** (x_t)
 $\downarrow$
explore again around x_t
improved? yes $\Rightarrow$ continue the
pattern
no $\Rightarrow$ step back and restart

Rule of thumb

P only decreases: do not reset P_0 at each success.

On Rosenbrock, resetting P_0 after each successful exploration: $n_f = 21\,816$ until $f < 10^{-4}$, against 430 ($\approx 50 \times$).

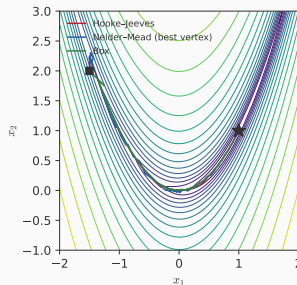
Data: x_0 , factor a , initial perturbations P_0 , tolerances T

```
1:  $P \leftarrow P_0$ ;  $x_b \leftarrow x_0$ 
2: loop ▷ start / restart
3:    $x_e \leftarrow$  exploration around  $x_b$ 
4:   if failed then
5:      $P \leftarrow P/2$ ; if all  $P_j < T_j$ : return  $x_b$ 
6:   else
7:     loop ▷ pattern moves
8:        $x_t \leftarrow x_b + a(x_e - x_b)$  ▷ trial point
9:        $x'_e \leftarrow$  exploration around  $x_t$ 
10:      if  $f(x'_e) \geq f(x_e)$  then  $x_b \leftarrow x_e$ ; exit (restart)
11:      else  $x_b \leftarrow x_e$ ;  $x_e \leftarrow x'_e$ 
12:    end if
13:  end loop
14: end if
15: end loop
```

Stopping: the exploration fails at all scales down to T , in *all* coordinates. Cost: each pattern move spends $1 + 2n$ evaluations (the trial point and the new exploration).

Where it fails, and the four methods on Rosenbrock

- **The stopping test can be fooled:** if a narrow valley passes between the points $\mathbf{x} \pm P_j \mathbf{e}_j$, the exploration fails along all axes although a descent direction exists. *Possible heuristic:* add a few random directions — it helps, but does not solve the problem of directional coverage.
- **Guarantee:** the coordinate exploration of HJ is the origin of the pattern search framework; formalized versions such as GPS (Torczon, 1997) make it possible to establish stationarity results under suitable assumptions (e.g. $f \in C^1$ and bounded level set $\Rightarrow \liminf \|\nabla f(\mathbf{x}_k)\| = 0$). With a finite step, stopping does not certify stationarity: it only says that none of the $2n$ directions improved with that step. HJ already uses a positive basis ($\pm \mathbf{e}_j$); modern pattern search generalizes it.
- In curved valleys it zigzags: successful pattern, failure, new exploration with a reduced step.



Rosenbrock from $(-1.5, 2)$ n_f to $f < 10^{-4}$	
Nelder-Mead (edges 0.5)	166
Hooke-Jeeves ($P_0 = 0.5, a = 2$)	430
localized random [†]	975 [867, 1052]
Box ($\Delta_0 = 1$)	13 445

[†] median [Q1, Q3], 30 runs. On this problem, with these parameters and this criterion.

On the computer: from Hooke–Jeeves to modern pattern search

The core, in MATLAB:

```
for j = 1:n                                % exploration (xe = copy of xb)
    xp = xe; xp(j) = xp(j) + P(j); fp = f(xp); % +P_j
    xm = xe; xm(j) = xm(j) - P(j); fm = f(xm); % -P_j
    if fp < fe && fp <= fm, xe = xp; fe = fp; % the best of three
    elseif fm < fe,      xe = xm; fe = fm; end
end
xt = xb + a*(xe - xb);                    % pattern: trial point
```

The full functions (Exploratory, HookeJeeves, with info.nfev) are in the course companion code.

In practice, the variables should be scaled and $P_{0,j}$ and T_j chosen relative to the scale of each variable (one may be in 10^{-3} m, another in 10^5 Pa).

patternsearch (MATLAB)

A modern pattern search (positive bases with $2n$ or $n + 1$ directions), which generalizes classical Hooke–Jeeves, with convergence theory (GPS) under suitable assumptions.

Hooke–Jeeves (the origin) → modern
pattern search
→ directional direct search
(poll, local component of GLODS, 5.1)
→ DMS, multiobjective (6.5)

Summary table — 3.2

Method	Information	Cost	Guarantee	Main limitation
Random (3.2.1)	values of f	1 per sample	global, in probability, as $N \rightarrow \infty$ (full support, f continuous)	$\ x - x^*\ \sim N^{-1/n}$
Grid (3.2.2)	values of f	m^n in total	$f_{\text{grid}} \rightarrow f^*$ (f continuous); $f_{\text{grid}} - f^* \leq L \frac{\sqrt{n}}{2} h$ (f L -Lipschitz)	exponential in n
Nelder–Mead (3.2.3)	values of f	1–2 per iteration ($n + 2$ if it shrinks)	no general one (McKinnon, 1998)	may stagnate
Box (3.2.4)	values of f	2^n per iteration	no general guarantee presented in this course; a small box does not certify stationarity at finite Δ	2^n ; only diagonals
Hooke–Jeeves (3.2.5)	values of f	$2n$ per exploration	formalized versions (GPS): stationarity in the limit, under suitable assumptions	axes; zigzag

random + grid = **explore**

Nelder–Mead + Box + Hooke–Jeeves = **refine locally**

NM has no general guarantee; HJ is the origin of pattern search, whose formalized versions (GPS) have stationarity in the limit under suitable assumptions (3.2.5); for Box we present no general guarantee. Only evaluations of f : the direction is inferred, and that costs evaluations; derivative-free $\neq$ robust to noise.

KEY TAKEAWAYS — 3.2.5

1. Hooke–Jeeves alternates axis-by-axis **exploration** ($\pm P_j$ in each coordinate, keep the best; $2n$ evaluations) and **pattern move** (extrapolating in the direction of progress).
2. The pattern **accelerates** and the exploration **corrects**: the path curves and follows oblique valleys, but zigzags in curved valleys.
3. The perturbations **P** only decrease, and only after an unsuccessful exploration.
4. The coordinate exploration of HJ is the origin of pattern search; formalized versions (GPS) have **stationarity in the limit, under suitable assumptions**; with a finite step, stopping does not certify stationarity.
5. Hooke–Jeeves $\rightarrow$ modern pattern search $\rightarrow$ GLODS poll (5.1) $\rightarrow$ DMS (6.5).

Next (3.3): so far we have inferred directions only through evaluations of f , and that costs evaluations. **What if f is smooth and we have ∇f ?** — derivative-based methods.

References and resources

References Full list: *Course bibliography*, at the end of the course PDF.

- Hooke, R., Jeeves, T. A., *“Direct search” solution of numerical and statistical problems*, Journal of the ACM 8(2):212–229, 1961.
- Torczon, V., *On the convergence of pattern search algorithms*, SIAM Journal on Optimization 7(1):1–25, 1997.
- Conn, A. R., Scheinberg, K., Vicente, L. N., *Introduction to Derivative-Free Optimization*, SIAM (MPS-SIAM Series on Optimization 8), 2009 — Ch. 7.
- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012 — Sec. 3.3.3 (the exploratory move used here).

Resources

- Companion code: Exploratory, HookeJeeves (`info.nfev`). MATLAB: `patternsearch` (Global Optimization Toolbox).